

>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

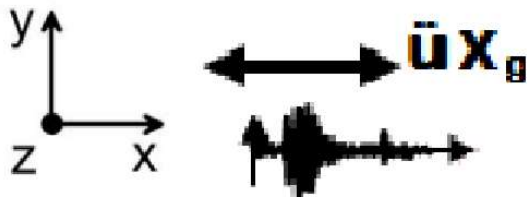
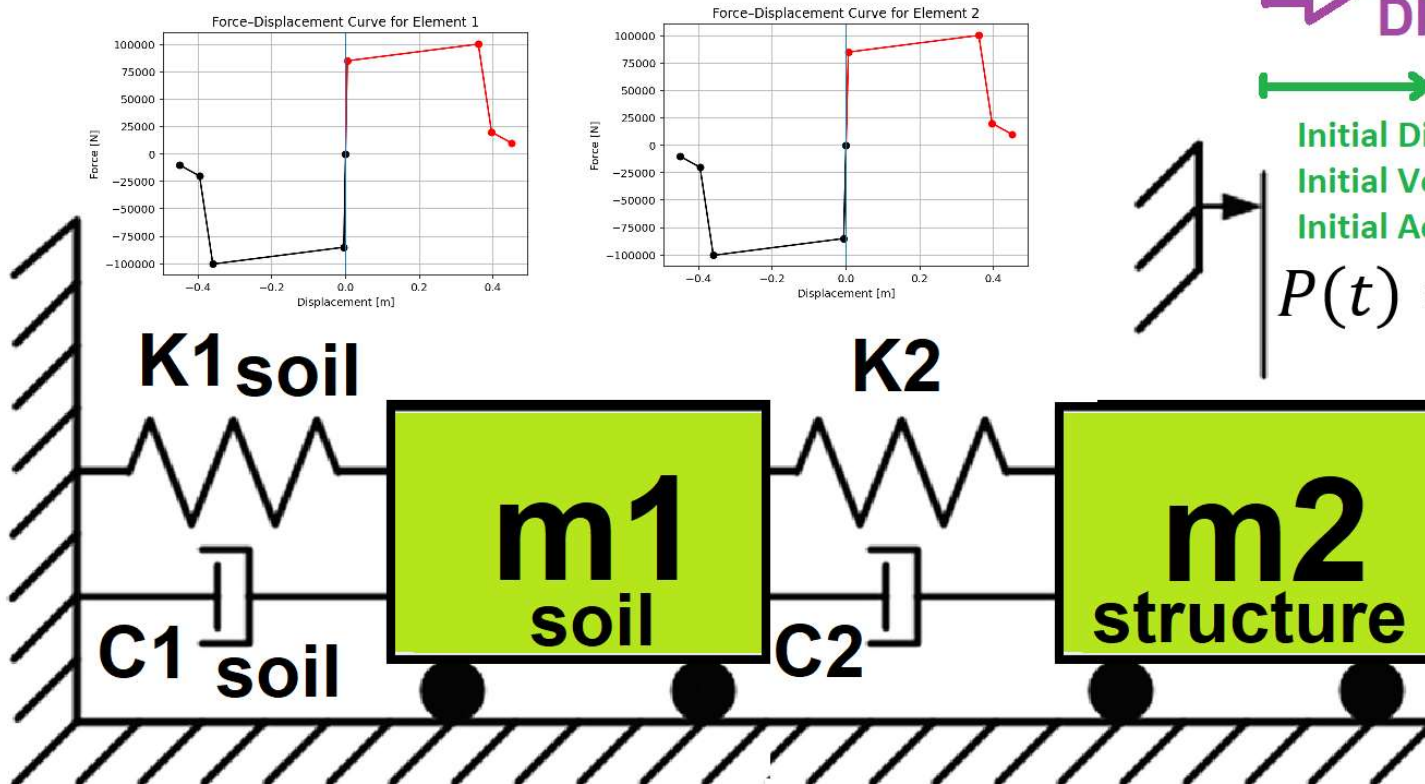
COMPREHENSIVE EVALUATION OF THE EQUIVALENT VISCOUS DAMPING RATIO OF A NONLINEAR MDOF (SOIL-STRUCTURE) SYSTEM UNDER MULTIPLE LOADING PROTOCOLS

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)

➡ INCREMENTAL
DISPLACEMENT

↔
Initial Displacement
Initial Velocity
Initial Acceleration

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$



Energy quantities and the damping formula

- **Dissipated energy** E_d – the area enclosed by the hysteresis loop. This is the net work done per cycle and represents the energy that must be removed from the system to return it to the original state. Physically, it corresponds to plastic yielding, friction, cracking, or other irreversible mechanisms.
- **Maximum strain energy** E_s – defined here as

$$E_s = \frac{1}{2} F_{\max} u_{\max}$$

where $u_{\max} = \max |u|$ is the peak absolute displacement in the loop and $F_{\max}$ is the absolute value of the force **at that same displacement**. This definition assumes a **secant stiffness** $k_{\text{sec}} = F_{\max}/u_{\max}$ and represents the elastic strain energy stored at peak displacement if the system were linear with that stiffness.

The equivalent viscous damping ratio is then:

$$\xi_{eq} = \frac{100 E_d}{4\pi E_s} \quad [\%]$$

This stems from equating E_d to the energy dissipated per cycle by a linear viscous damper, $E_d = \pi c \omega u_{\max}^2$, while $E_s = \frac{1}{2} k u_{\max}^2$; substituting $\xi = c/(2m\omega)$ and $\omega^2 = k/m$ yields the familiar expression $\xi = E_d/(4\pi E_s)$.

$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

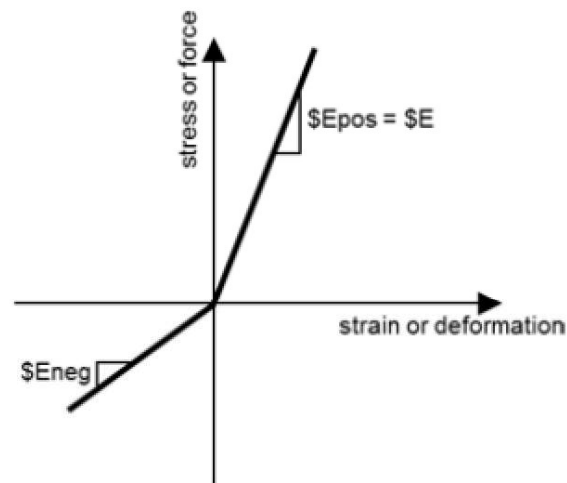
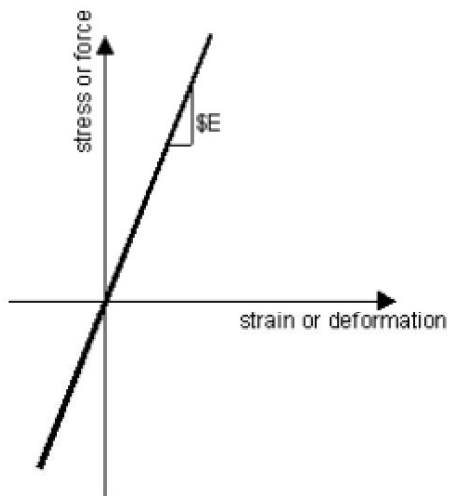
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

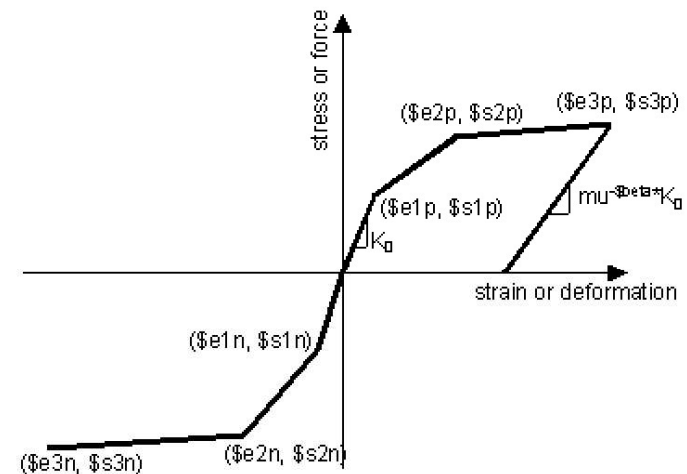
Δ_u = Lateral Ultimate Displaement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

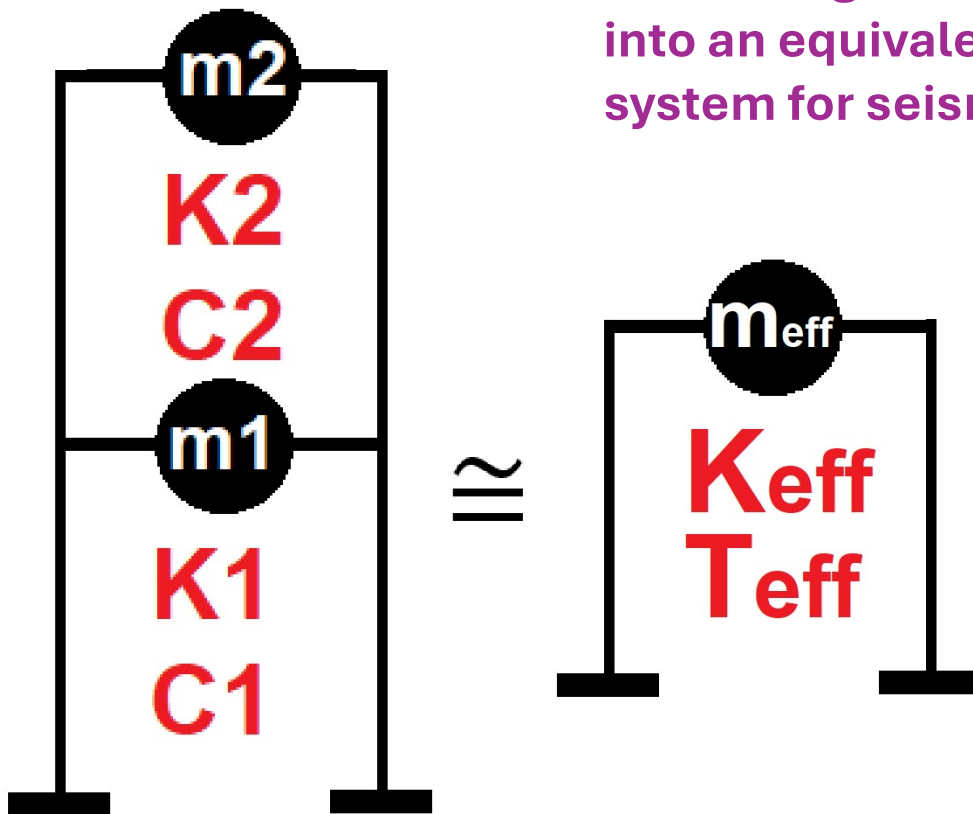


ELASTIC MATERIAL



INELASTIC MATERIAL

Displacement-based seismic analysis transformation, converting a multi-degree-of-freedom (MDOF) system into an equivalent single-degree-of-freedom (SDOF) system for seismic assessment



$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i} \quad m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d} \quad T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

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C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...RE_MDOF_8_ANA\COMPUTE_EFFECTIVE_PROPERTIES_FUN.py

P(t)_SOIL_STRUCTURE_MDOF_8_ANA.py X COMPUTE_EFFECTIVE_PROPERTIES_FUN.py X

```
1  """ EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED PUSHOVER ANALYSIS
2  # Change MDOF to SDOF System with Displacement Based Design Concept
3  # THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
4  """
5  This script implements a displacement-based pushover transformation,
6  converting a multi-degree-of-freedom (MDOF) system into an equivalent
7  single-degree-of-freedom (SDOF) system for seismic assessment.
8  It calculates effective modal properties-displacement, mass, and
9  stiffness-by weighting element forces and nodal displacements according
10 to a presumed deformed shape.
11 The derived effective period provides a simplified dynamic characteristic
12 for performance-based engineering. The visualizations effectively track the
13 evolution of these equivalent parameters throughout the nonlinear analysis steps.
14 """
15 #-----
16 displacement_X_1 = np.array(list(node_displacements.values())[1]) # DOF 02
17 displacement_X_2 = np.array(list(node_displacements.values())[2]) # DOF 03
18
19
20 ele_force_01 = np.array(list(ele_force.values())[0]) # ELEMENT 01
21 ele_force_02 = np.array(list(ele_force.values())[1]) # ELEMENT 02
22
23 STIFF_X_1 = np.abs(ele_force_01 / displacement_X_1)
24 STIFF_X_2 = np.abs(ele_force_02 / displacement_X_2)
25
26 MX2 = (MASS[0] * np.square(displacement_X_1) +
27        MASS[1] * np.square(displacement_X_2))
28
29 MX = (MASS[0] * np.array(displacement_X_1) +
30       MASS[1] * np.array(displacement_X_2))
31 EFFECTIVE_DISP_X = MX2 / np.abs(MX)
32
33 EFFECTIVE_MASS_X = np.abs(MX) / EFFECTIVE_DISP_X
34
35
```

Effective Displacement - Median: 0.16588

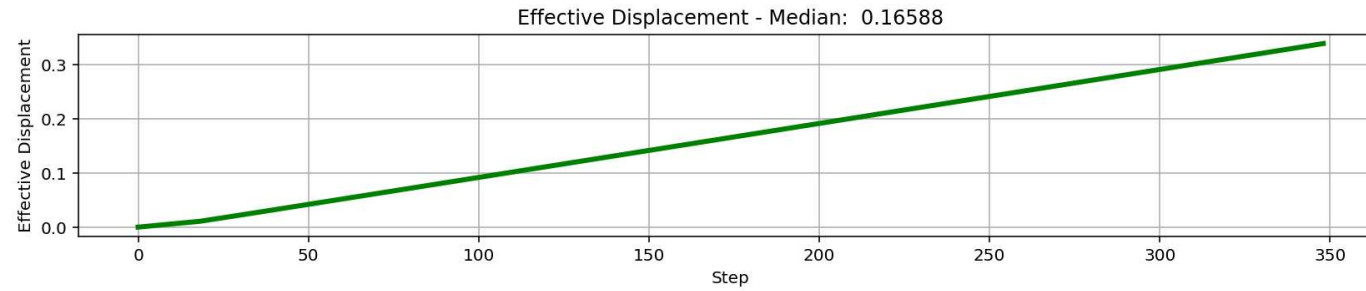
Effective Mass - Median: 4998.29864

Effective Stiffness - Median: 829950.70654

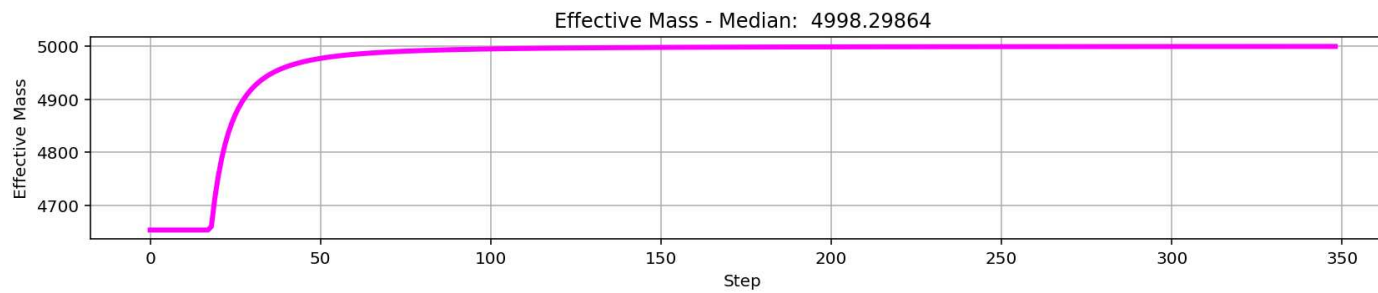
Effective Period - Median: 0.48768

IPython Console Files Help Variable Explorer Debugger Plots History

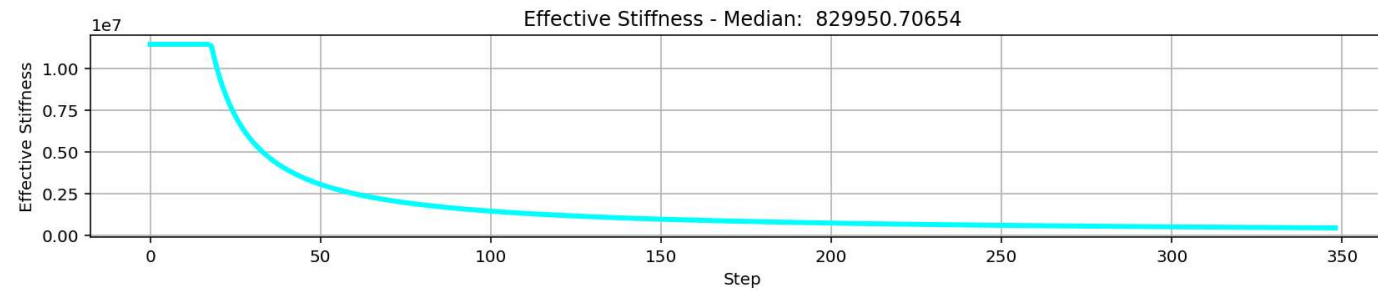
Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 56%



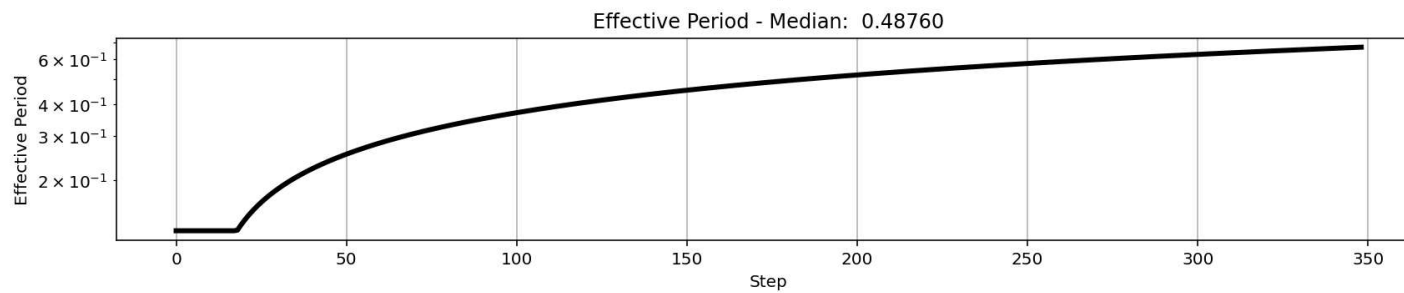
$$\Delta_d = \frac{\sum_{i=1}^n m_i \Delta_i^2}{\sum_{i=1}^n m_i \Delta_i}$$



$$m_{eff} = \frac{\sum_{i=1}^n m_i \Delta_i}{\Delta_d}$$



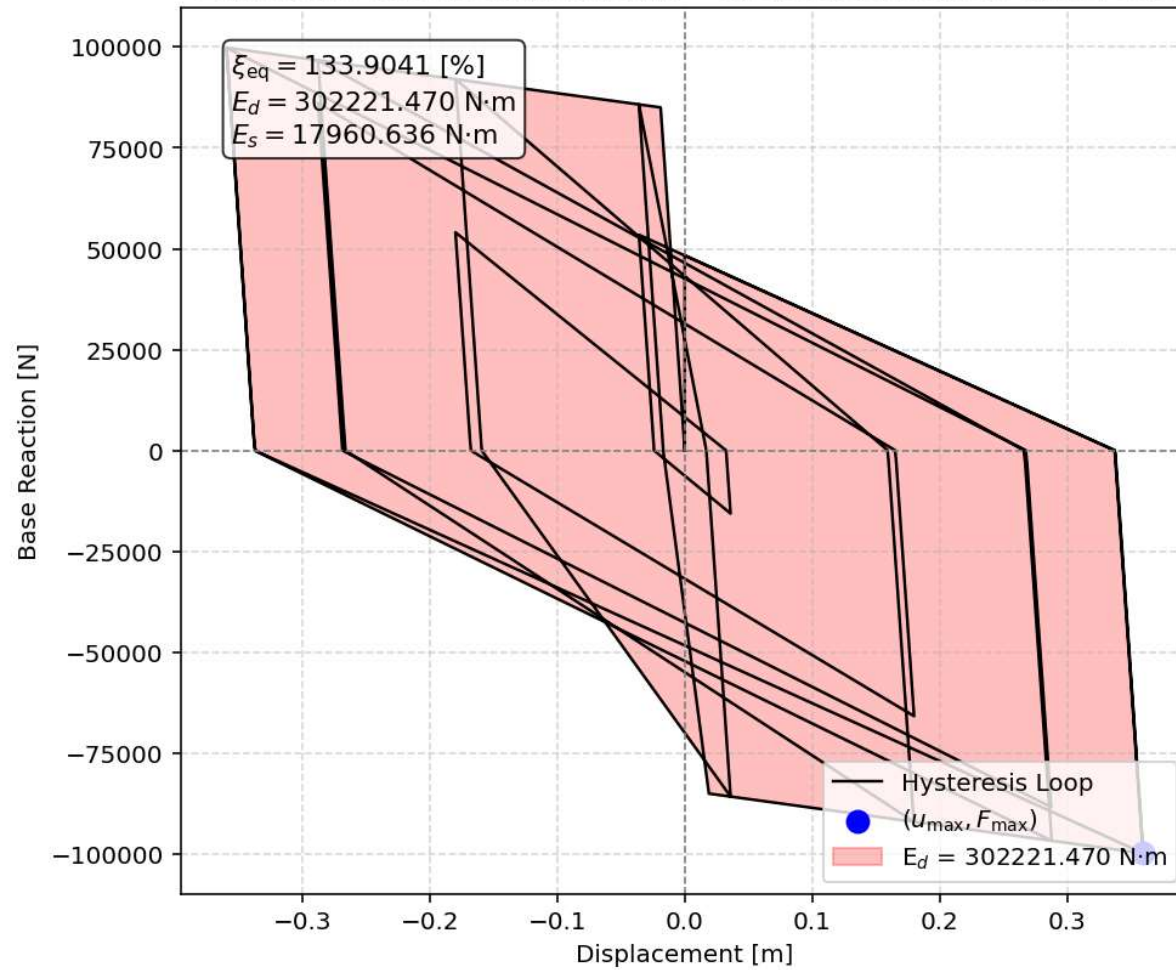
$$k_{eff} = \frac{\sum_{i=1}^n k_i \Delta_i}{\Delta_d}$$



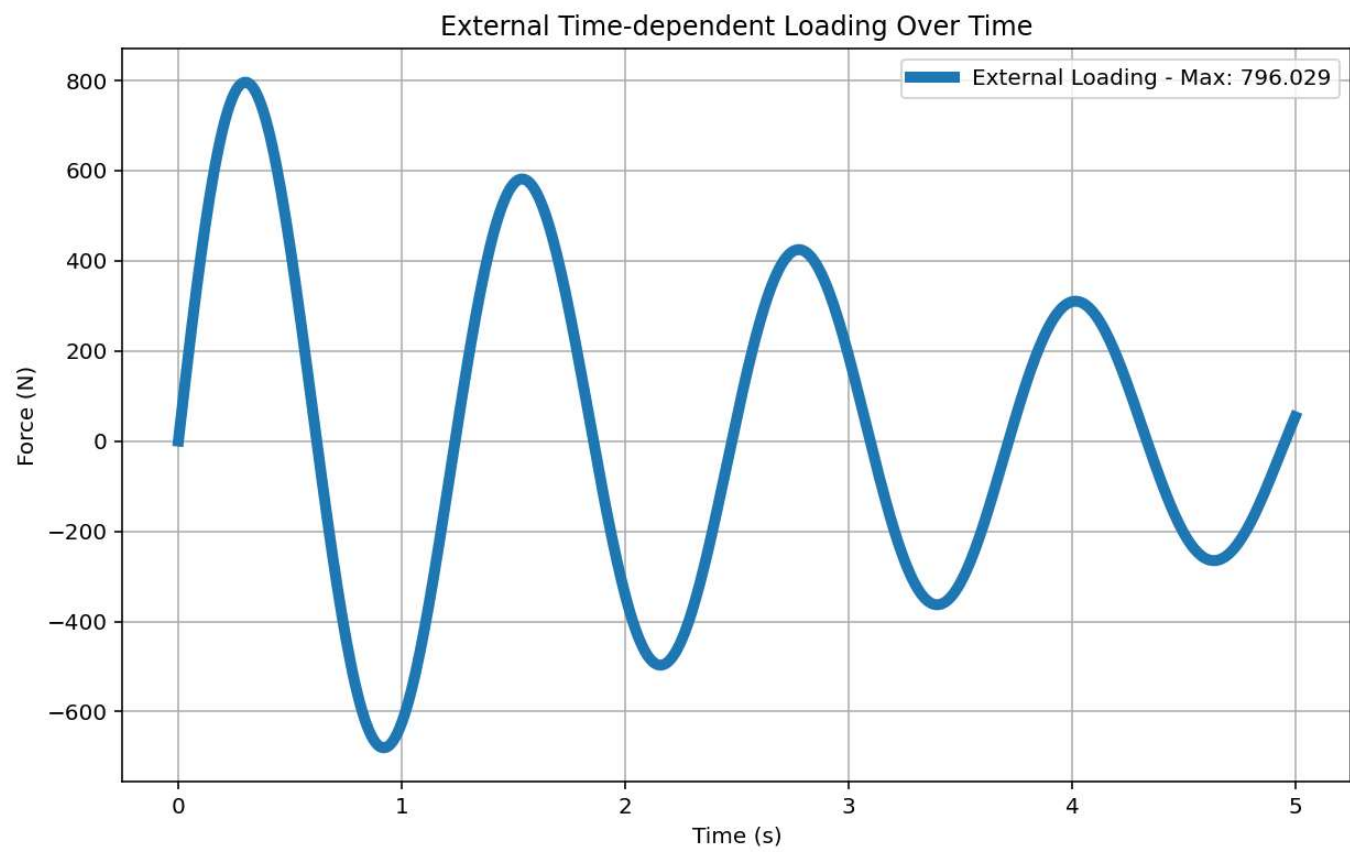
$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

CYCLIC DISPLACEMENT ANALYSIS

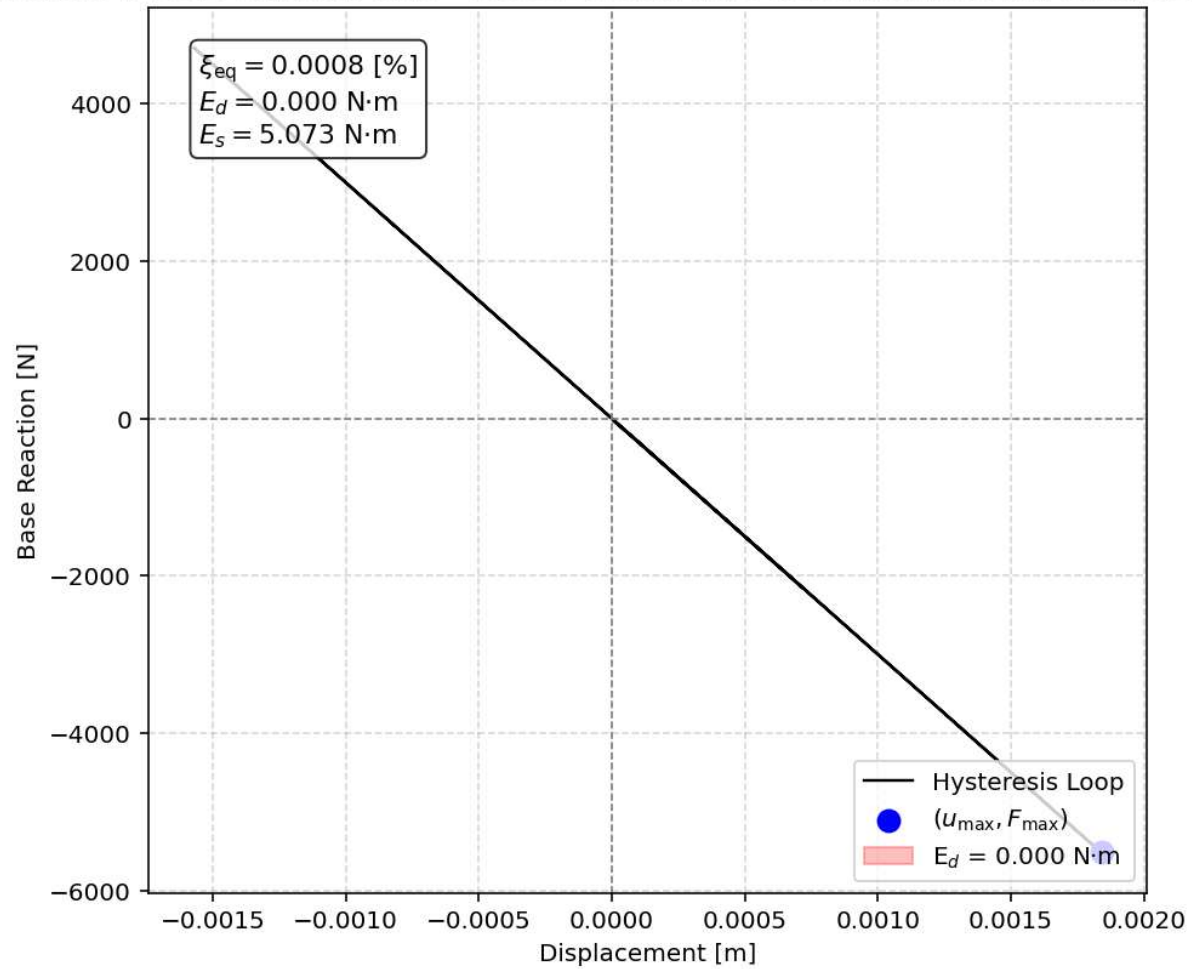
Equivalent viscous damping ratio - Cyclic Displacement Hysteresis



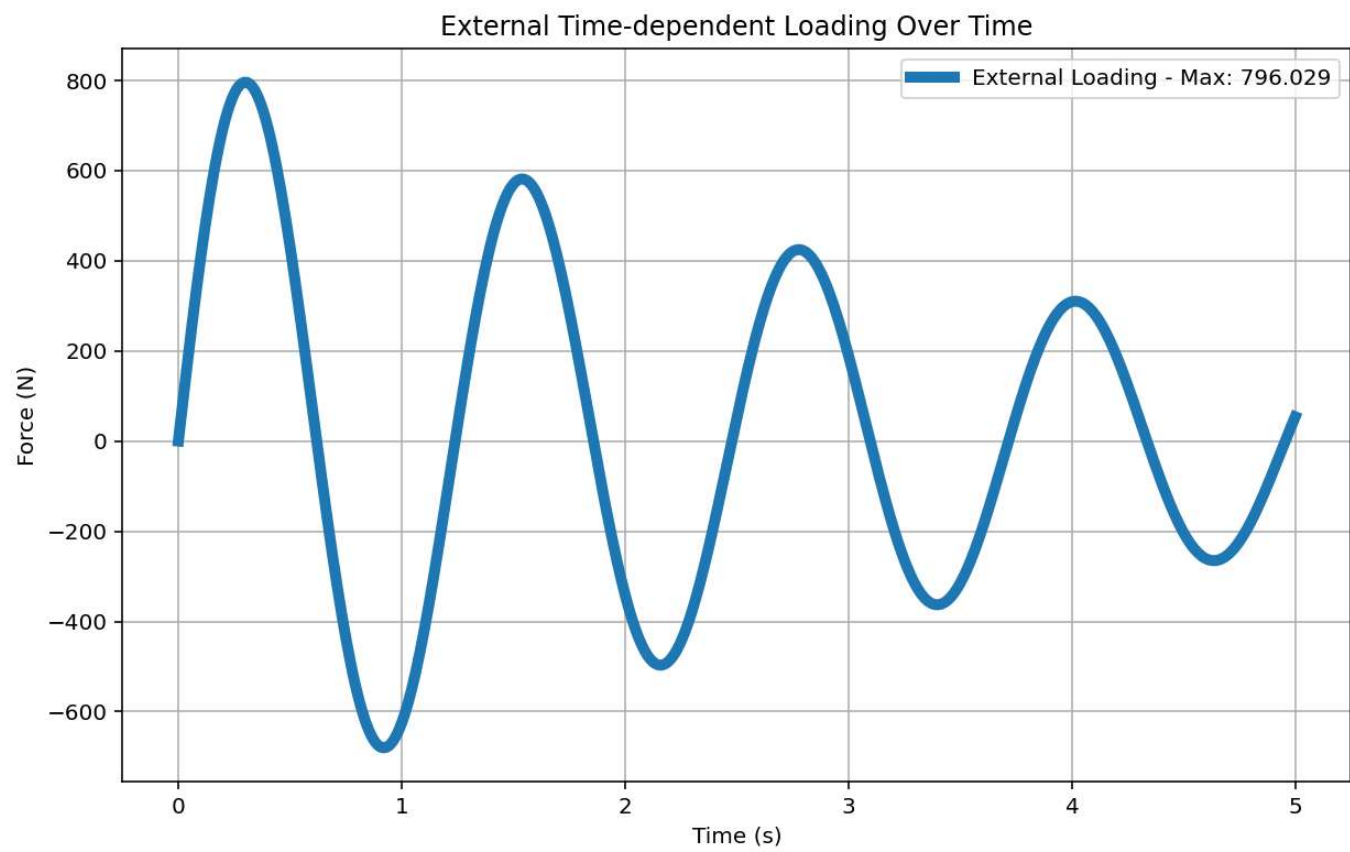
STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS



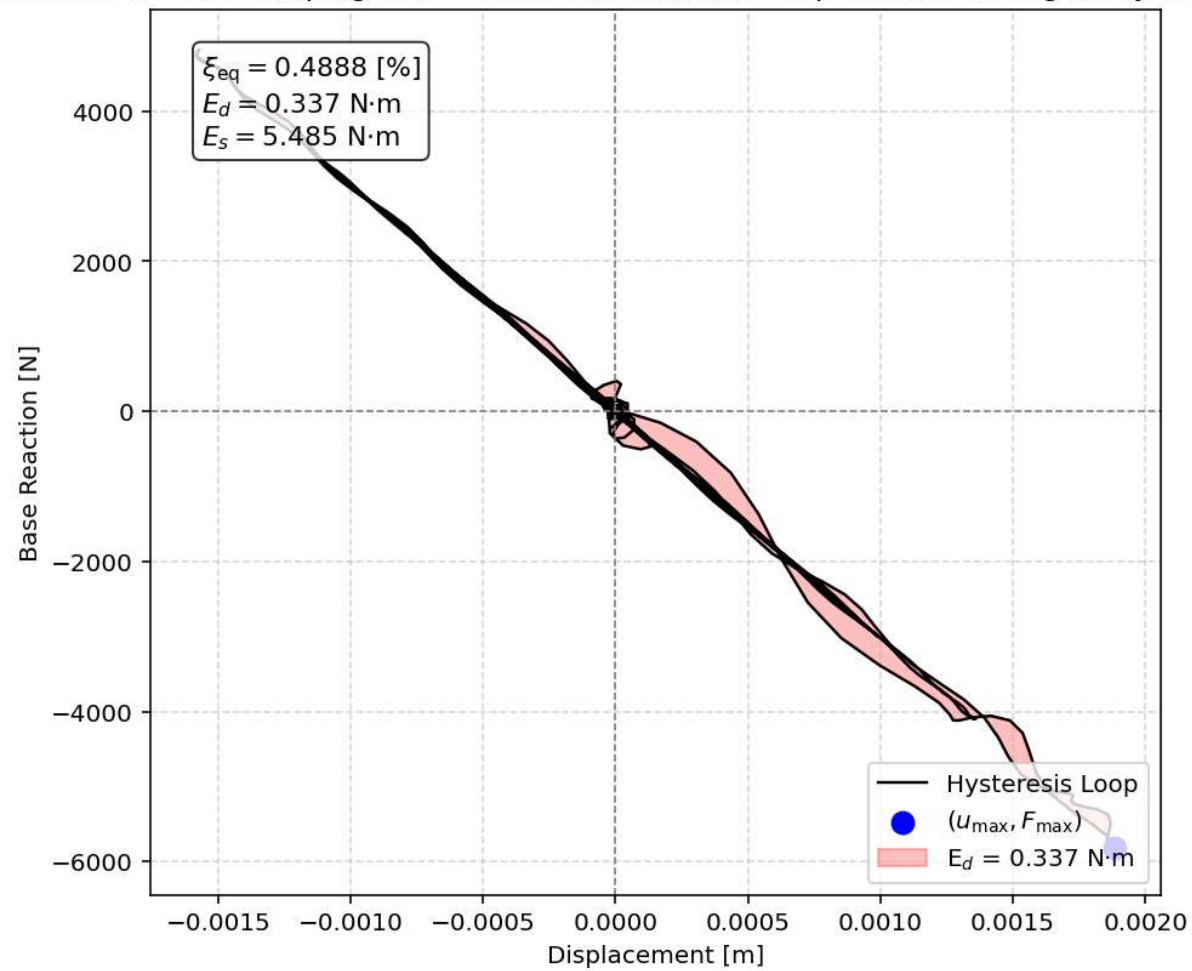
Equivalent viscous damping ratio - Static External Time-dependent Loading Analysis Hysteresis



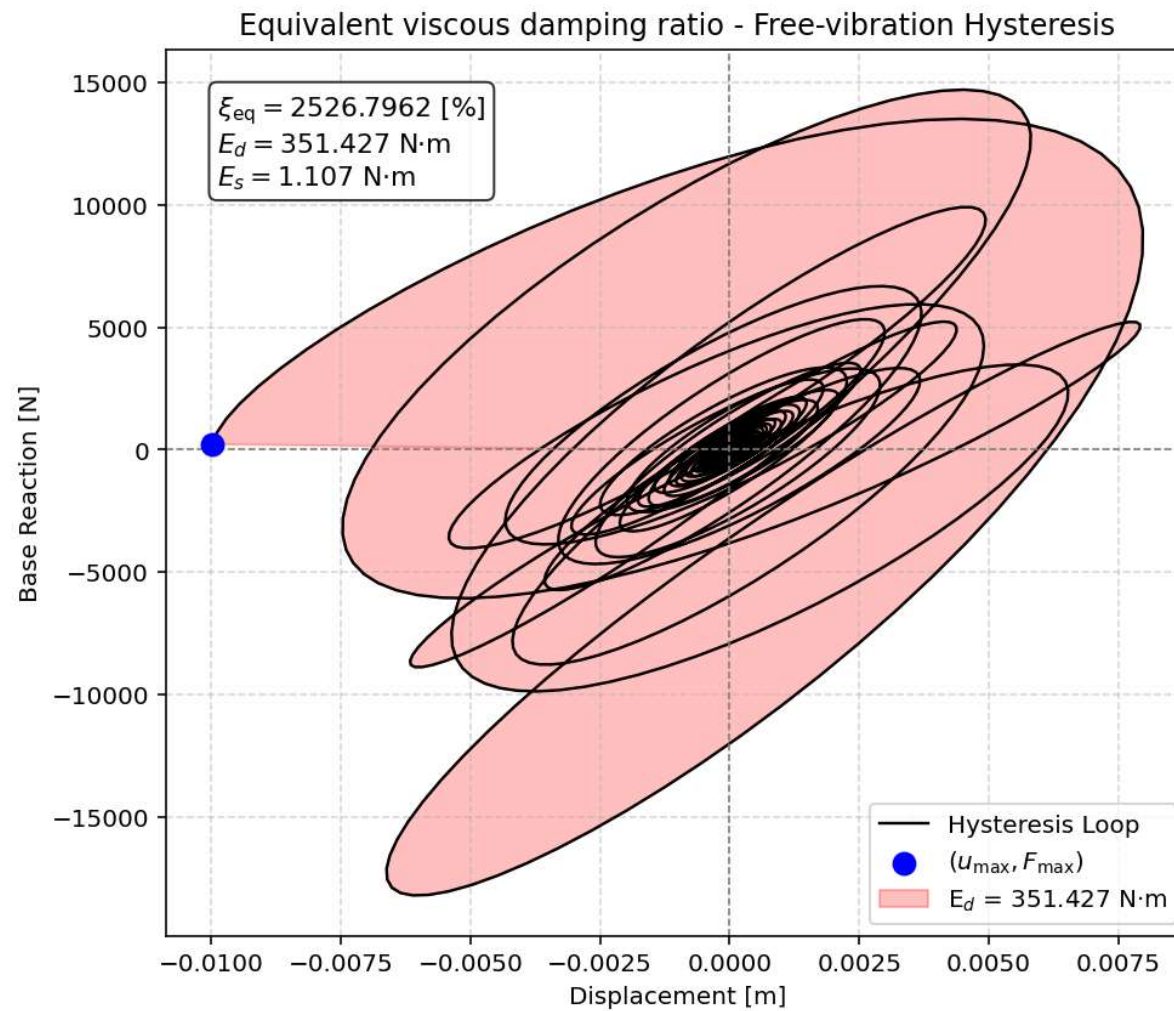
DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS



Equivalent viscous damping ratio - Static External Time-dependent Loading Analysis Hysteresis



FREE-VIBRATION ANALYSIS



SEISMIC ANALYSIS

Equivalent viscous damping ratio - Seismic Hysteresis

